greatlearning
Power Ahead

Capstone Presentation

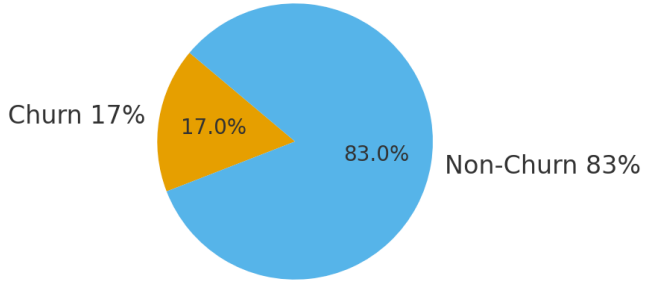
Guidelines to be followed :

- **Presentation should complete in 10 mins**
- **10 minutes will be devoted to Q&A**

Business Problem Understanding

Business Problem

The company is facing rising customer churn as a result of competitive pressure, increasing customer expectations, and service dissatisfaction. Churn directly impacts recurring revenue because losing one client means losing multiple linked accounts.



Constraint	Description
No churn detection system	⚠️ Manual processes make early identification difficult.
Large & complex dataset	📊 Behavior + interactions require structured modeling.
Multiple interacting factors	🔄 Complaints, service score, tenure combine to influence churn.
Need early prediction	⏳ Intervention must happen before disengagement.

Scope

Analyze customer behavior and service interaction data to identify churn drivers and develop a predictive model for early retention intervention.

Objectives:

- 🔍 Detect churn early to support proactive decisions.
- 📈 Identify key churn drivers such as complaints, service experience, and tenure.

Modeling Approach Used & Why

Multiple model analysis:

Several prediction methods were tested to compare how well each one identifies customers at risk.
Different models learn different patterns
Testing multiple options ensures reliability.
Accuracy, recall and business impact were the elements included in the analysis.

Models used

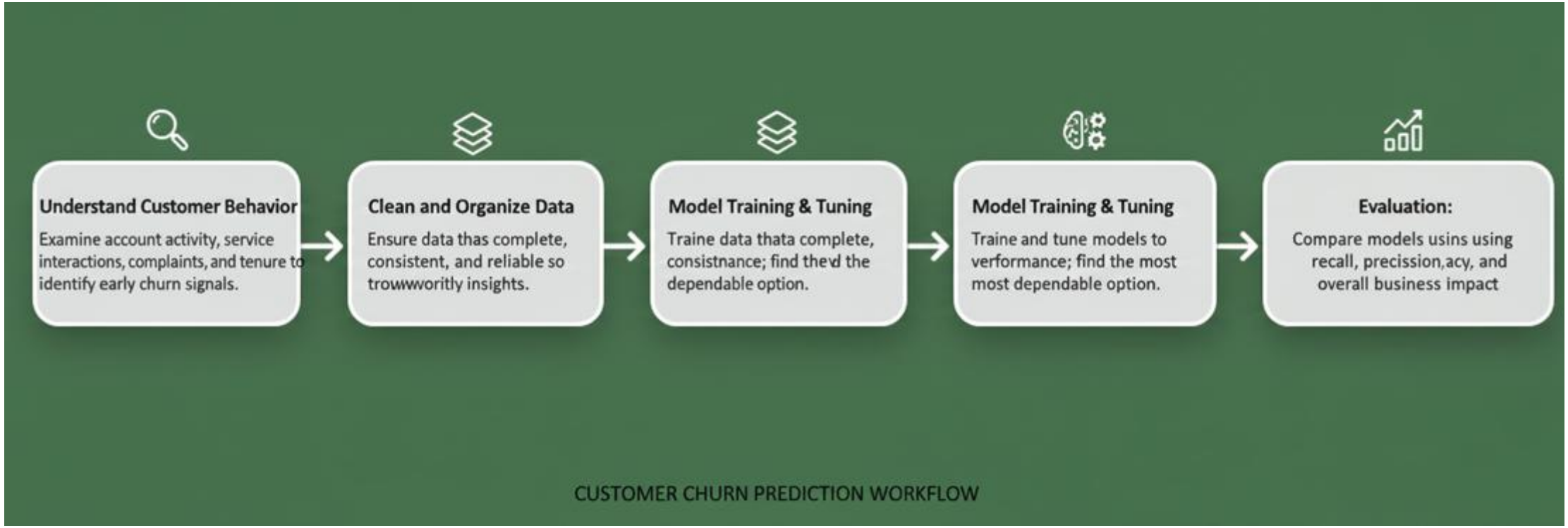
Model	Business Description
KNN	Simple pattern-matching model that compares customers to similar past cases.
Decision Tree	Learns easy-to-understand decision rules based on customer behavior.
Random Forest	Combines many decision trees to create more stable and reliable predictions.
XGBoost	Advanced model that captures complex patterns and interactions for the strongest early warnings.

Metrics used

Metric	Description
Recall	% customers who actually churned
Precision	% of customers flagged as churners that were actually churners
Accuracy	Overall correctnes, or % of all predictions that model got right.
XGBoost	0.874

Modeling Approach Used & Why

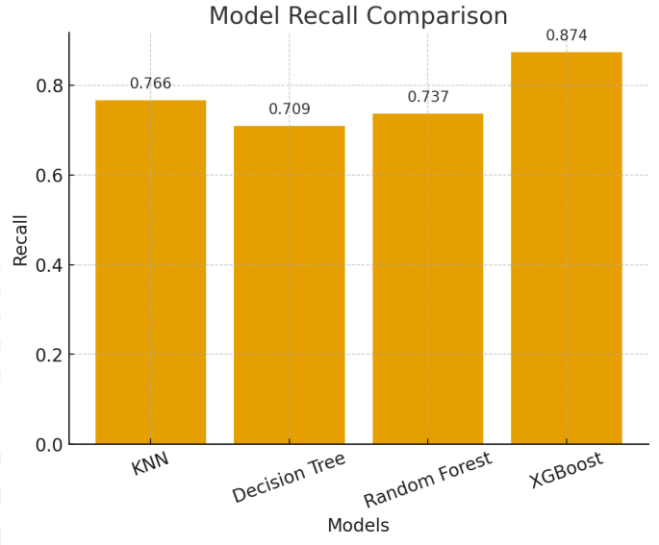
Modeling Process



Insights from Analysis

Model performance

Model	Recall (How many real churners we catch)	Precision (How many flagged customers are truly at risk)	Accuracy (Overall correctness)	Conclusion
KNN	**0.766**	**0.884**	**0.944**	❌ **Too risky** Misses many churners — high revenue leakage
Decision Tree	**0.709**	**0.788**	**0.919**	❌ **Unstable** Very unstable — fails with real customers
Random Forest	**0.737**	**0.936**	**0.947**	⚠️ **Good but not best** Reliable but not the safest for churn detection
XGBoost	**0.874**	**0.953**	**0.972**	✓ **Best model** Maximizes customer retention — highest business value

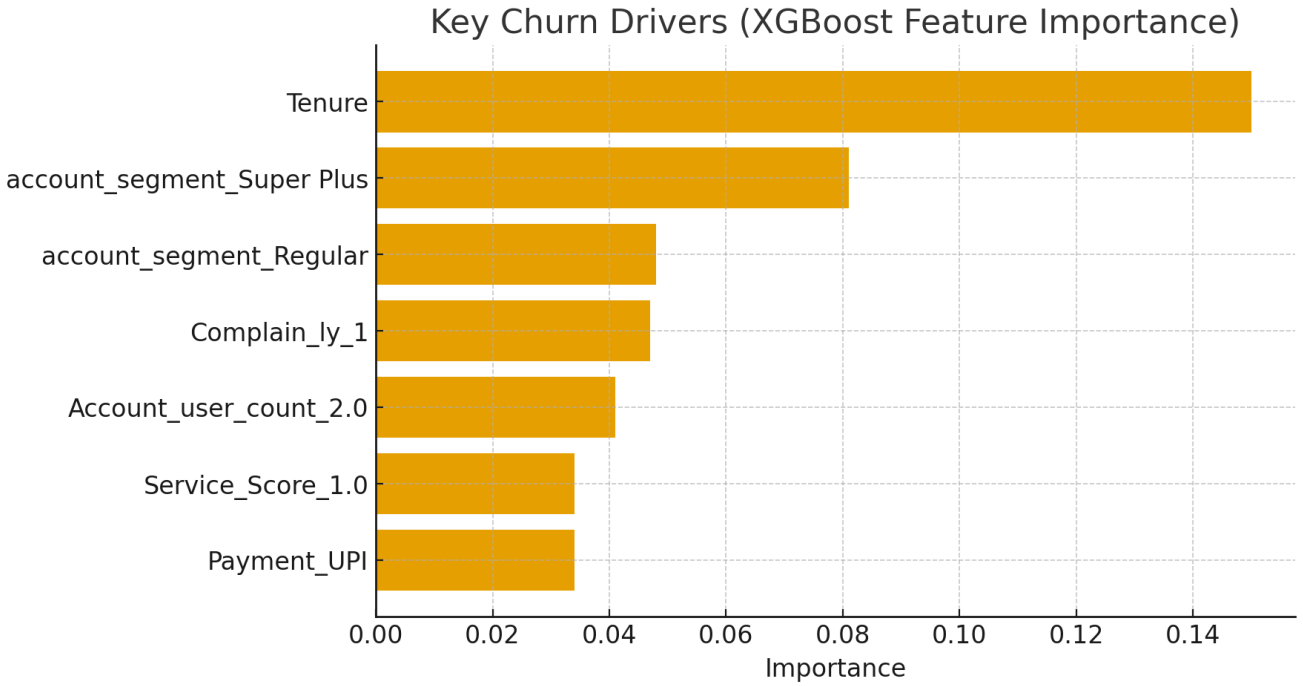


- XGBoost detects the highest number of churn-risk customers early
- It generates minimal false alarms
- Recall is prioritized because missing churners is far more costly than contacting a few extra customers.

Insights from Analysis

Key Churn Drivers

- Tenure and customer segment are the strongest predictors of churn, followed by complaints and service-related interactions



Tenure is the top churn driver, with premium segments and complaint history also playing significant roles.

Recommendations

- Strengthen Onboarding for New Users
  - Proactive support program to combat short tenure churn.
- Segment-Specific Retention Strategies
  - Tailor benefits (value-adds, loyalty) based on specific segment risks.
- Proactive Complaint Management
  - Use a 'red flag' system; ensure swift resolution and follow-up.
- Improve Multi-User Account Experience
  - Optimize UX/communication for 2-user accounts.
- Address Low Service Scores Quickly
  - Initiate immediate outreach to resolve issues and improve service.
- Optimize Payment Options
  - Ensure seamless payment and incentivize stickier methods.
- Enhance Customer Care Efficacy
  - Focus on reducing wait times and improving first-contact resolution.